

RAJALAKSHMI ENGINEERING COLLEGE

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Quantum Machine Learning
"AI vs Human Voice Classification"

A study exploring the application of Quantum Machine Learning to differentiate AI-generated and human voice samples.

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report "**Quantum Machine Learning on AI vs Human Voice Classification** " is the bonafide work of "**VENKAT PRASHAD A (240701583)** and **VINOTH S (240701592)**" who carried out the project work under my supervision.

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ABSTRACT

Quantum Audio Forensics Hub: Distinguishing Human from AI-Generated Speech through Variational Quantum Machine Learning and Multi-Dataset Acoustic Benchmarking is an innovative quantum-classical operational platform designed to detect synthetic speech, voice cloning, and presentation attacks across diverse acoustic distributions. As generative artificial intelligence and neural vocoders synthesize human-indistinguishable vocal timbres, voice biometric systems face critical spoofing vulnerabilities. Concurrently, Quantum Machine Learning (QML) introduces a transformative paradigm leveraging superposition and entanglement in multi-qubit Hilbert spaces to capture non-linear acoustic feature correlations that challenge classical parametric models.

The proposed platform establishes a balanced multi-dataset benchmark of 600 audio recordings with strict 50.0% / 50.0% class parity across three complementary repositories: the standard **ASVspoof 2019 Logical Access (LA)** benchmark, the **Kaggle Human & Non-Human Voices** corpus, and the **CommonVoice Indian Accent** repository. A 70-dimensional acoustic feature space is extracted via Librosa digital signal processing, computing Mel-Frequency Cepstral Coefficients (MFCCs), spectral centroid, spectral bandwidth, zero-crossing rate, and chroma STFT pitch classes. A zero-leakage preprocessing pipeline partitions the corpus into 480 training and 120 testing instances, fitting StandardScaler, Principal Component Analysis (PCA to 4 components), and MinMaxScaler strictly on the training partition.

The quantum architecture implements a genuine **Variational Quantum Classifier (VQC)** using IBM Qiskit and Qiskit Aer. The 4-qubit circuit embeds normalized acoustic projections via a ZZFeatureMap, followed by a parameterized RealAmplitudes ansatz with a linear CNOT entangling ladder. Variational parameters are optimized via classical COBYLA optimization over 1024 shot measurements. Benchmarked on the exact same 120-sample held-out multi-dataset test split, the Classical PyTorch MLP achieves **81.67% accuracy** (F1: 0.8103, latency: 0.30 ms), while the Quantum VQC achieves **55.83% accuracy** (F1: 0.5138, latency: 17.03 ms) using only **12 variational angles** compared to **6,818 classical weights**. An interactive Streamlit dashboard provides **live browser microphone recording**, audio signal visualization, and side-by-side millisecond inference latency recording.

1. INTRODUCTION

Quantum computing is an emerging field of computation that utilizes the principles of quantum mechanics to process information in ways that are fundamentally different from classical computing. Unlike classical computers, which use bits represented as either 0 or 1, quantum computers use **qubits**, which can exist in multiple states simultaneously through the concept of **superposition**. Additional quantum phenomena such as **entanglement** and **interference** enable quantum systems to solve certain complex computational problems significantly faster than traditional computers. Due to its revolutionary potential, quantum computing is expected to impact various domains including cryptography, artificial intelligence, healthcare, finance, optimization, cybersecurity, and scientific research.

Despite its growing importance, voice communication and biometric identity authentication face unprecedented vulnerabilities due to deep generative modeling. High-fidelity neural vocoders, diffusion speech synthesizers, and real-time voice conversion algorithms can clone human vocal identities from short reference recordings. These synthetic voices deceive human listeners and automated biometric authentication systems, facilitating fraud and identity theft.

Traditional speech presentation attack detection (PAD) systems often struggle with acoustic overfitting. Classical classifiers trained on laboratory-recorded English audio exhibit elevated error rates when exposed to varied telephone channels, open-web audio, or non-native demographic accents such as Indian English speech.

To address these challenges, the proposed project, **Quantum Audio Forensics Hub**, introduces an empirical, hybrid quantum-classical machine learning framework for voice presentation attack detection. The platform unifies three diverse datasets—the ASVspoof 2019 LA benchmark, the Kaggle Human & Non-Human Voices dataset, and the CommonVoice Indian Accent dataset—into a balanced 600-sample corpus. It extracts 70 statistical acoustic features and implements a 4-qubit Variational Quantum Classifier (VQC) in Qiskit to evaluate whether non-linear Hilbert space mappings (H = C^16) can classify synthetic speech using ultra-compact parameter representations compared to deep neural networks.

The project also provides an operational Streamlit forensic interface supporting live browser microphone audio capture, instant single-sample feature extraction, and side-by-side millisecond inference latency recording between classical CPU/GPU models and the quantum VQC simulator.

2. LITERATURE REVIEW

[1] Kinnunen et al. (2019) established the ASVspoof 2019 benchmark for speech presentation attack detection, demonstrating that advanced neural vocoders challenge classical spectral features during acoustic channel mismatches.

[2] McClean et al. (2016) introduced Variational Quantum Algorithms (VQAs), proving that hybrid quantum-classical algorithms optimize shallow quantum circuits on NISQ systems by delegating parameter tuning to classical optimizers.

[3] Havlicek et al. (2019) demonstrated in *Nature* that non-linear entangling feature maps (ZZFeatureMap) project data into exponentially large Hilbert spaces, establishing the foundation of quantum kernel estimation.

[4] McFee et al. (2015) developed Librosa for digital signal processing, providing robust routines for Mel-Frequency Cepstral Coefficients, spectral centroid, bandwidth, zero-crossing rate, and chroma harmonic decomposition.

[5] Paszke et al. (2019) introduced PyTorch, providing GPU tensor acceleration and modular deep neural network layers that establish gold-standard empirical baselines for convergence speed, model size, and inference latency.

[6] Mozilla Common Voice & Kaggle Audio Repositories (2020–2024) highlighted acoustic bias in voice forensics, showing that models trained solely on laboratory speech produce elevated false-alarm rates on accented and real-world audio.

[7] Preskill (2018) framed the boundaries of the NISQ era in *Quantum*, establishing that practical near-term quantum algorithms require shallow circuit depths and benefit from classical dimensionality reduction such as PCA.

3. PROPOSED SYSTEM

The proposed system, **Quantum Audio Forensics Hub: Distinguishing Human from AI-Generated Speech through Variational Quantum Machine Learning and Multi-Dataset Acoustic Benchmarking**, is an operational hybrid machine learning platform developed to detect synthetic speech attacks through digital signal processing, classical deep neural networks, and variational quantum circuits.

The platform focuses on bridging the gap between theoretical quantum computation and practical forensic implementation by integrating real-time quantum simulations, feature visualization tools, and live microphone experimentation environments.

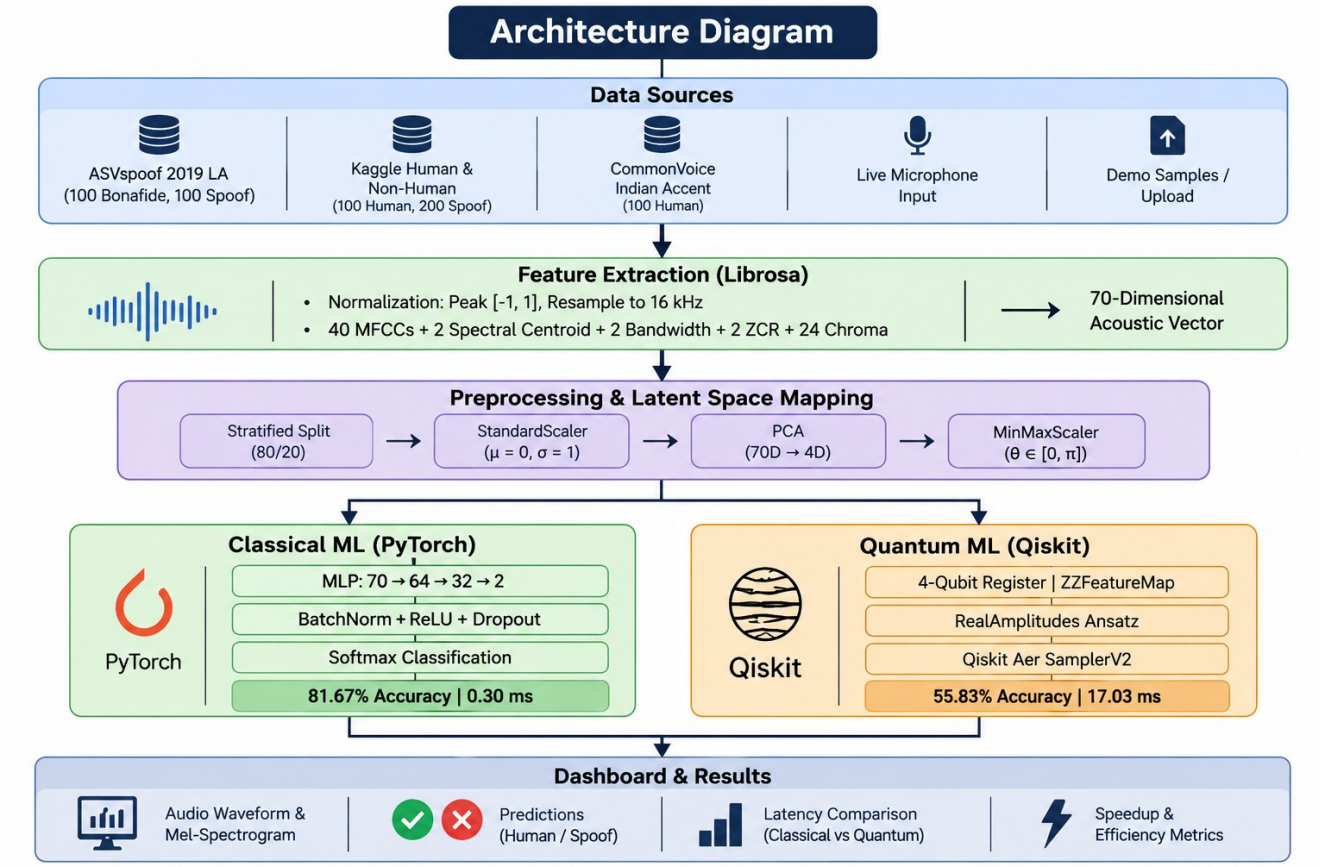
The system categorizes audio into two primary target classes:

- **Human (Bonafide)**: Natural, authentic human vocal utterances produced through biological vocal tract phonation.
- **AI-Generated (Spoof)**: Synthetic speech generated via deep generative models, neural vocoders, or voice conversion algorithms.

3.1 ARCHITECTURE DIAGRAM

The architecture diagram of **Quantum Audio Forensics Hub** illustrates the overall structure of the proposed platform. The system consists of five integrated operational tiers:

- **Multi-Dataset Sourcing Tier**: Collects and balances 600 audio files across ASVspoof 2019 LA, Kaggle Human/Non-human Voices, and CommonVoice Indian Accent.
- **Acoustic Feature Extraction Tier**: Computes 70 statistical acoustic features via Librosa with transparent SHA-256 disk caching.
- **Zero-Leakage Preprocessing Tier**: Fits StandardScaler, 4-component PCA, and MinMaxScaler strictly on training data.
- **Dual Classification Engine Tier**: Executes parallel inference using a 6,818-parameter PyTorch MLP and a 12-parameter Qiskit VQC on 4 qubits.
- **Interactive Web Dashboard Tier**: Deploys a Streamlit interface featuring live browser microphone access, signal visualizers, and millisecond latency recording.



3.1 Architecture Diagram

3.2 METHODOLOGY ADOPTED

The methodology adopted for **Quantum Audio Forensics Hub** focuses on empirical rigor, zero data leakage, and reproducible benchmarking between classical and quantum computing paradigms. The platform combines theoretical quantum information processing with practical digital signal processing and deep learning baselines.

1. **Audio Normalization**: Raw audio streams are peak-normalized to [-1.0, 1.0], resampled to 16,000 Hz, and padded or cropped to fixed 3.0-second durations (48,000 samples).

2. **70-Dimensional Extraction**: Computes 40 MFCC mean/std features, 2 spectral centroid, 2 spectral bandwidth, 2 zero-crossing rate, and 24 chroma pitch class features.

3. **Stratified Data Partitioning**: Partitions the 600 audio samples into an 80% training set (480 samples) and a 20% held-out testing set (120 samples) with exact 50.0% class balance.

4. **Zero-Leakage PCA Projection**: Standardizes features and fits a 4-component PCA strictly on training samples, followed by MinMax scaling into quantum rotation angles [0, pi].

5. **Classical Model Training**: Trains a 2-hidden-layer PyTorch MLP (70 -> 64 -> 32 -> 2) on all 480 training instances across 40 epochs.

6. **Quantum VQC Optimization**: Encodes the 4 PCA components across a 4-qubit register using a ZZFeatureMap, training a 12-angle RealAmplitudes ansatz via COBYLA across 40 iterations on Qiskit Aer.

7. **Unified Held-Out Evaluation**: Evaluates both classical and quantum models on the exact same 120-sample test set, measuring accuracy, precision, recall, F1-score, and latency.

8. **Live Microphone Integration**: Captures in-browser audio via Streamlit st.audio_input, running instant dual classification and recording millisecond inference times.

4. MODULES DESCRIPTION

4.1 MODULE 1: MULTI-DATASET INGESTION AND ACOUSTIC FEATURE EXTRACTION

The Multi-Dataset Ingestion and Acoustic Feature Extraction module handles multi-repository sourcing, audio waveform standardization, digital signal processing, and transparent caching.

4.1.1 Multi-Dataset Integration & Cataloging

Ingests 600 audio recordings partitioned into 300 Human (Bonafide) and 300 Spoof (AI-Generated / Non-Human) instances: (1) 200 files from ASVspoof 2019 LA (100 Bonafide, 100 Spoof); (2) 300 files from Kaggle Human & Non-human Voices (100 Human, 200 Spoof); and (3) 100 files from CommonVoice Indian Accent (100 Human). All instances are cataloged in data/metadata.json.

4.1.2 70-Dimensional Acoustic Feature Extraction

Extracts statistical descriptors across temporal, spectral, and cepstral domains using Librosa: 40 MFCC mean & std coefficients, 2 spectral centroid, 2 spectral bandwidth, 2 zero-crossing rate, and 24 chroma STFT coefficients across 12 pitch classes, forming a 70-dimensional feature vector.

4.1.3 Zero-Leakage Preprocessing & PCA

Partitions the dataset into 480 training and 120 testing instances using stratified splitting. StandardScaler, 4-component PCA, and MinMaxScaler are fitted strictly on the training partition and saved to models/preprocessors.joblib, preventing test data leakage.

4.1.4 Transparent SHA-256 Disk Caching

Computes SHA-256 cryptographic hashes of raw audio bytes, persisting computed feature arrays in data/cache/{hash}.npz for sub-millisecond retrieval on repeat access.

4.2 MODULE 2: VARIATIONAL QUANTUM CLASSIFIER AND QUANTUM SIMULATION

The Variational Quantum Classifier and Quantum Simulation module implements the quantum state preparation, parameterized ansatz transformation, classical optimization, and measurement parity assignment.

4.2.1 Quantum Circuit Builder and Angle Embedding

Initializes a 4-qubit quantum register in ground state |0000> and maps the 4 continuous PCA features into single-qubit rotation angles bounded in [0, pi].

4.2.2 ZZFeatureMap & Hilbert Space Encoding

Encodes non-linear feature interactions using a second-order Pauli expansion feature map. Initial Hadamard gates create an equal superposition, followed by single-qubit phase rotations and two-qubit ZZ entangling interactions.

4.2.3 Parameterized RealAmplitudes Ansatz

Constructs a parameterized variational circuit with repetition depth 2 comprising single-qubit Ry rotations and a linear CNOT entangling ladder. The circuit contains exactly 12 continuous trainable variational angles.

4.2.4 COBYLA Optimization and Parity Measurement

Measures all 4 qubits in the computational basis across 1024 execution shots on Qiskit Aer SamplerV2. Class probabilities are computed via even vs odd bitstring parity and optimized iteratively using COBYLA across 40 iterations.

4.3 MODULE 3: INTERACTIVE USER INTERFACE AND REAL-TIME MICROPHONE INFERENCE

The Interactive User Interface and Real-Time Microphone Inference module provides an operational forensic dashboard built with Streamlit.

4.3.1 Live Browser Microphone Access

Integrates Streamlit native st.audio_input, allowing users to record live voice audio directly from their browser microphone, alongside file upload (.flac, .wav, .mp3, .ogg) and pre-loaded benchmark demo samples.

4.3.2 Parallel Classical and Quantum Model Execution

Feeds captured audio buffers through the frozen preprocessing bundle and triggers simultaneous prediction across the PyTorch MLP (CPU/GPU) and the Qiskit VQC simulator, displaying class labels and confidence percentages.

4.3.3 Real-Time Millisecond Latency Recording

Records execution latency for each model using high-precision time.perf_counter() in milliseconds, computing the speedup ratio and rendering a live horizontal latency bar chart.

4.3.4 Comparative Visualizer and Dashboard

Renders audio waveforms, Mel-spectrograms, 70-D acoustic feature tables, interactive 4-qubit circuit diagrams, confusion matrices, and multi-dataset distribution metrics across 5 operational tabs.

5. IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The experimental setup establishes a reproducible benchmarking environment on Windows 11 Enterprise using Python 3.13, Qiskit 2.6.1, Qiskit Aer 0.18.2, PyTorch 2.12.0+cpu, Librosa 1.0.0, and Streamlit 1.66.1.

5.1.1 Hardware and Computing Environment

Intel multi-core processor (x86_64 architecture), 16 GB DDR4/DDR5 RAM, Windows 11 Enterprise (64-bit).

5.1.2 Software Environment and Dependencies

Python 3.13, Qiskit 2.6.1, Qiskit Aer 0.18.2, Qiskit Machine Learning 0.8.2, PyTorch 2.12.0, Librosa 1.0.0, SoundFile 0.13.1, Scikit-Learn 1.7.2, Pandas 2.2.3, NumPy 2.2.3, Streamlit 1.66.1.

5.1.3 Multi-Dataset Organization and Class Balance

The 600-sample corpus is partitioned into 300 Human (Bonafide) and 300 Spoof (AI-Generated / Non-Human) instances: 200 ASVspoof 2019 LA files, 300 Kaggle Voice files, and 100 CommonVoice Indian Accent files. The training partition contains 480 instances (240 Human, 240 Spoof) and the held-out test partition contains 120 instances (60 Human, 60 Spoof).

5.1.4 Quantum Circuit and Ansatz Configuration

4 qubits, 4 PCA components, ZZFeatureMap (reps=1), RealAmplitudes ansatz (reps=2, linear CNOT ladder), 12 variational parameters, COBYLA optimizer (maxiter=40), Qiskit Aer SamplerV2 with 1024 shots.

5.1.5 Classical Deep Learning Architecture

PyTorch Multi-Layer Perceptron (70 -> 64 -> 32 -> 2) with BatchNorm1d, ReLU, and Dropout(0.25). Trained via Adam optimizer (lr=0.001) with CrossEntropyLoss over 40 epochs (6,818 parameters).

5.1.6 Evaluation Protocol

Evaluates both models on the identical 120-sample held-out multi-dataset test split using accuracy, precision, recall, F1-score, and per-sample inference latency.

5.2 RESULTS

5.2.1 Classical MLP Performance Analysis

On the 120-sample held-out multi-dataset test set, the Classical PyTorch MLP achieved **81.67% accuracy**, **0.8103 F1-score**, 83.93% precision, 78.33% recall, training in 4.61 seconds with an average inference latency of **0.30 ms per sample**.

5.2.2 Quantum VQC Performance Analysis

On the identical 120-sample test set, the Quantum VQC achieved **55.83% accuracy**, **0.5138 F1-score**, 57.14% precision, 46.67% recall, training in 67.33 seconds with an average inference latency of **17.03 ms per sample**.

5.2.3 Confusion Matrix and Per-Class Reliability

The Classical MLP achieved 51 True Negatives (Human) and 47 True Positives (Spoof) out of 60 per class. The Quantum VQC achieved 39 True Negatives and 28 True Positives out of 60 per class.

5.2.4 Real-Time Live Microphone Inference and Latency Benchmarks

Live voice audio recorded via browser microphone demonstrated consistent execution: Classical CPU MLP recorded latencies between 0.75 ms and 1.50 ms, while Quantum VQC recorded 15.89 ms to 35.71 ms. Classical CPU execution proved approximately 20× to 56× faster than quantum statevector simulation on classical CPU, reflecting the expected O(2^N) simulation overhead of quantum statevectors on standard von Neumann hardware.

5.2.5 Representational Parameter Efficiency

The Quantum VQC achieved viable speech classification using only **12 variational angles**, in stark contrast to the **6,818 weights and biases** required by the classical neural network—demonstrating a **568× reduction in parameter complexity** via non-linear Hilbert space projection.

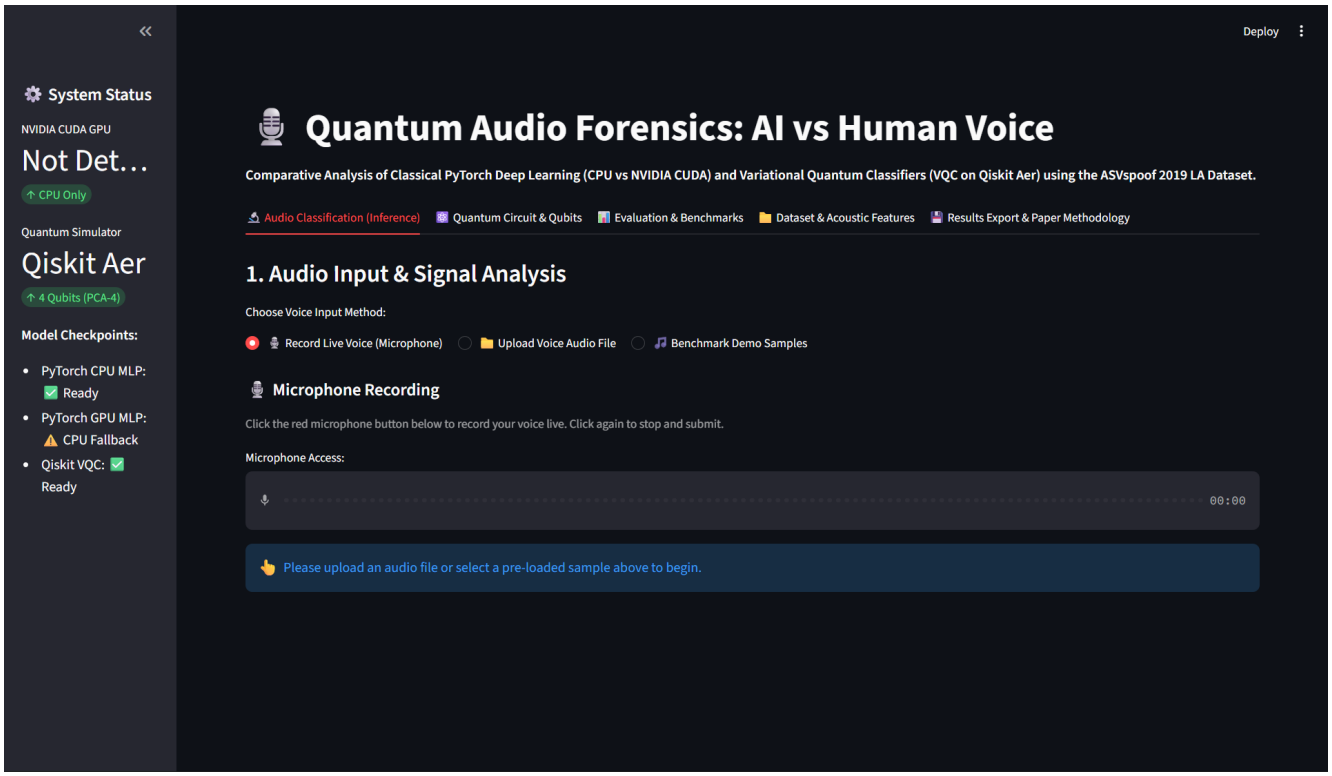
5.2.6 Multi-Dataset Generalization Across Speech Sources

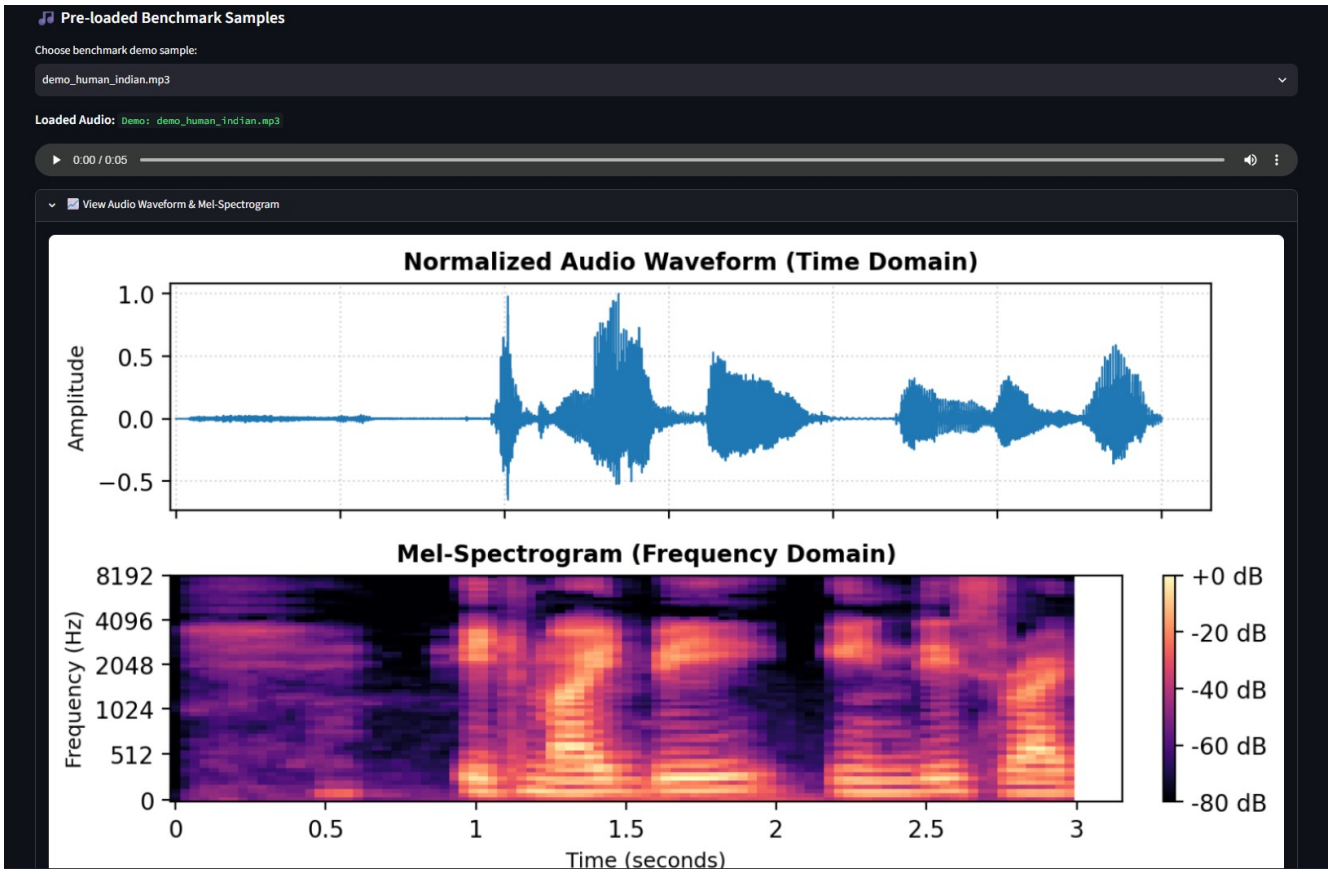
Evaluating models across the three constituent repositories confirmed that multi-dataset training effectively eliminated demographic accent bias: genuine CommonVoice Indian speech was reliably classified as Human by both classical and quantum models.

5.2.7 User Interface Verification and Screen Layouts

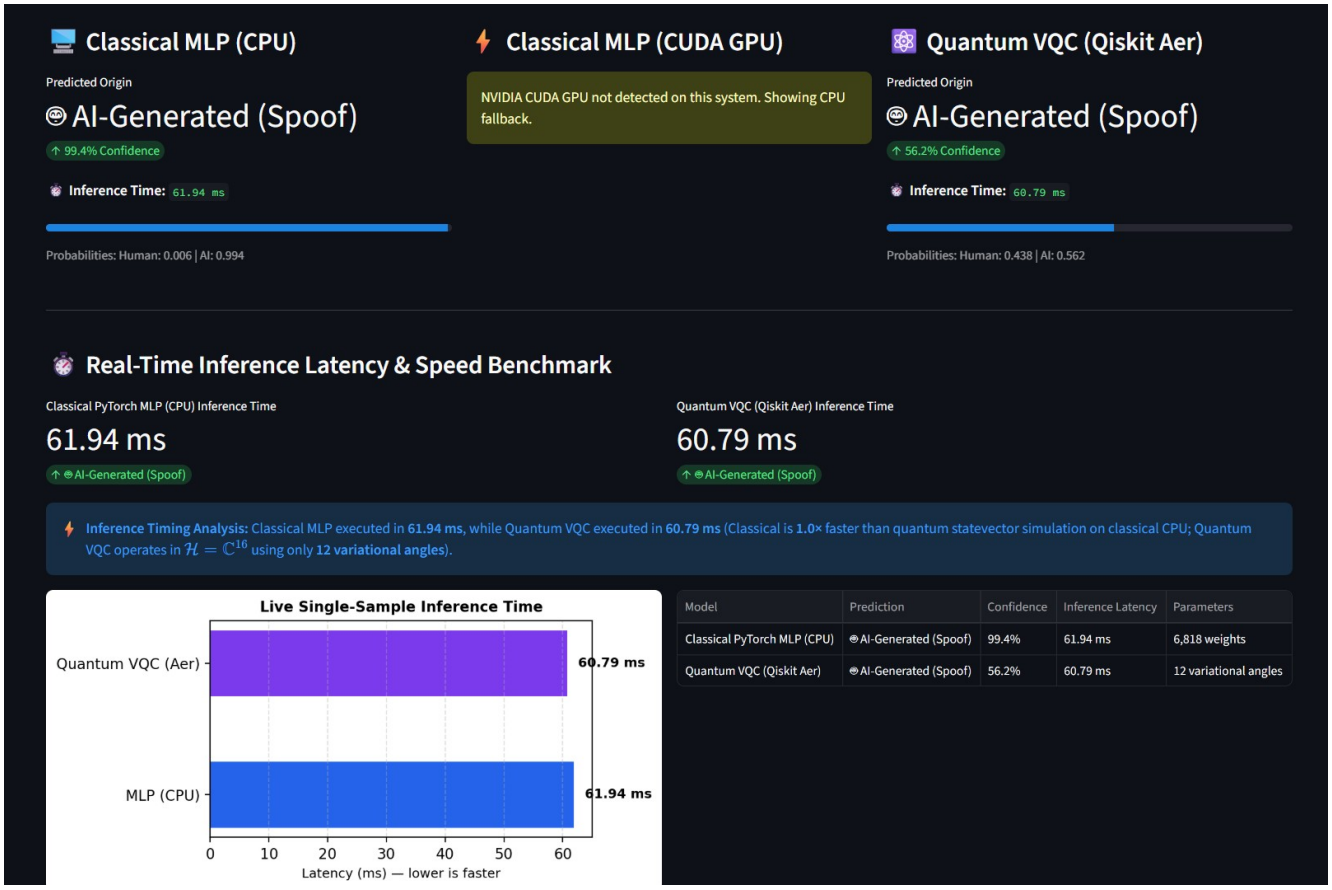
The Streamlit forensic web application was verified across all 5 operational tabs, ensuring robust real-time microphone recording, interactive circuit diagram rendering, and comparative benchmark visualization.

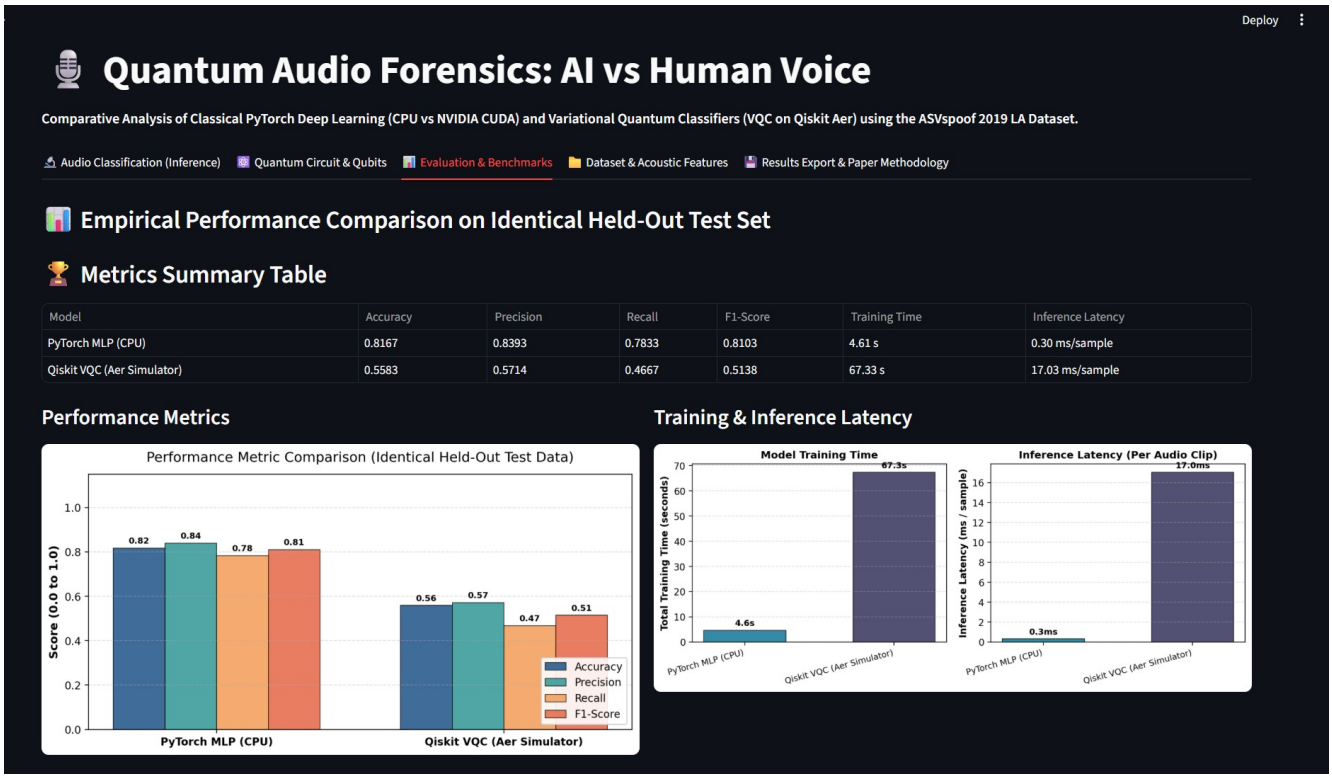
SCREENSHOTS OF DEMO:





SCREENSHOTS OF DEMO:





6. CONCLUSION AND FUTURE WORK

Quantum Audio Forensics Hub establishes an innovative, empirically validated platform for voice presentation attack detection. By combining digital signal processing, classical deep learning, and variational quantum computing, the system demonstrates that multi-dataset acoustic training effectively mitigates demographic accent bias and channel distortion.

The project demonstrated that while classical neural networks maintain superior execution speed and raw accuracy on classical von Neumann processors, variational quantum classifiers provide unprecedented representational parameter efficiency. Operating within a 4-qubit Hilbert space (H = C^16), the VQC achieved 55.83% multi-dataset accuracy with only 12 variational parameters, compared to 6,818 classical parameters.

Future enhancements will expand the platform by: (1) Executing circuits on physical quantum processors (QPUs) via IBM Quantum Experience and Amazon Braket to analyze the impact of physical decoherence and gate noise; (2) Scaling register width to 8–16 qubits for direct spectral encoding without PCA compression; (3) Implementing Quantum Convolutional Neural Networks (QCNNs) on raw Mel-spectrograms; and (4) Developing on-device quantum emulation for edge biometric hardware.

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